

as used to make power semiconductors. Winky-blinky LEDs have a junction that is encapsulated by an insulating, epoxy lens. This leaves only the leads to conduct heat away from the junction. In contrast, power LEDs, like other semiconductor devices, are manufactured on a chip. This chip is then fixed to a heat slug that penetrates through the package, and the connections are made to external terminals by bonding wires. The slug is then encapsulated with a silicone gel and covered with a hard plastic lens that is coated with phosphorus. This encapsulation method avoids stress on the bonding wires. The epoxy lens will not be practical, because of thermal expansion.

Heat is the biggest enemy of power LEDs. However, the lifetime of LEDs can exceed 50,000 hours, in comparison to 8000 hours of fluorescent bulbs and 2000 hours of incandescent bulbs. To achieve such a long lifetime, the junction temperature of the LED must be kept low. The actual temperature limit is a subject of debate among major LED manufacturers, but, in general, the junction temperature must be kept at 80°C or less to achieve a long lifetime.

When run continuously at temperatures above 80°C, the LED can fail in less than 10,000 hours. At temperatures near 80°C, the light output falls off rapidly in the first 10,000 hours. But, thereafter, the LED continues to generate a reduced light output for long. At more moderate temperature levels, the LED produces a relatively consistent light output over its entire lifetime.

Although LEDs have evolved to become very efficient sources of light, every LED system design is a tradeoff between light output, efficacy and heat-sink design. It may be necessary to drive a power LED at a reduced power level to meet temperature and heat-sink design requirements. Furthermore, packaging requirements of the lighting fixture might limit the ability to provide good heat-sinking.

The most common method for dimming in large installations is analogue 0-10V DC control. It can be incorporated into the design using an analogue-to-digital converter (ADC) channel.

Power LEDs with power levels exceeding 3W have become widely available. However, it is still easier to meet thermal design requirements using multiple smaller LEDs in the 1-2W power range. Furthermore, greater efficacy can be obtained when the LED is driven at a lower current. LED system designs will continue to become more practical as the efficacy of the LED is increased.

How to drive a power LED?

LEDs require a constant-current source, rather than constant-voltage. For the winky-blinky and lower-power types, a resistor does the trick. For LEDs above 1W, the resistor becomes impractical. Standard switch-mode power supply (SMPS) topologies and controllers can be used to drive the LEDs at these high power levels, using LED current instead of voltage as the feedback to the controller. The choice of topology depends on the system input voltage, LED forward voltage and the number of LEDs connected in series.

Clean power

So far we have discussed only the power-conversion circuits that allow an LED or fluorescent tube to be driven from a DC bus. How current is drawn from the AC line is of equal importance. Typically, a bridge rectifier circuit with filter capacitors consumes current from the AC line only at the peaks of the AC input voltage. The result is a current waveform with a high harmonic content and poor power factor.

An active PFC circuit can improve conversion of AC power into DC power. In simple terms, an active PFC circuit forces the current consumption of the circuit to track the envelope of the incoming AC line voltage. PFC helps to meet energy-efficiency re-

quirements. Thus it helps customers achieve a faster payback for electronic lighting controls.

Active PFC is most easily implemented using a voltage-boost circuit with an outer voltage-feedback loop and an inner current-control loop. The voltage-feedback control loop provides the demand for the inner current-control loop and determines whether more or less current is required to achieve the desired bus voltage. The current loop demand is then used to scale a sinusoidal reference signal.

The sinusoidal reference signal can be derived in two ways. First, it can be measured directly from the rectified AC input voltage. A simpler solution uses a sinusoidal reference stored in the memory of the MCU. A PWM channel on the MCU can be used as a simple digital-to-analogue converter to generate the sinusoidal reference. An R-C filter is connected to the PWM output pin to convert the PWM signal into a voltage.

The values in the sinusoidal table are scaled up or down by the current demand from the voltage control loop. The sinusoidal reference signal stored in the memory is synchronised to the zero crossings of the incoming AC voltage, using an interrupt on the MCU.

The use of a sinusoidal reference table assumes that the incoming AC voltage is purely sinusoidal and does not have any distortion. This assumption is practical for many applications. Consequently, the method is known as indirect PFC. Fig. 7 shows the block diagram of indirect PFC.

Communication control and status

MCUs allow communication to be integrated into lighting applications. This is especially important for commercial